FINAL YEAR PROJECT

PROJECT PROPOSAL

BS Information Technology — 7th Semester

BLOCKCHAIN-BASED CERTIFICATE VERIFICATION SYSTEM

STUDENT & SUPERVISOR DETAILS

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PROJECT PROPOSAL

Blockchain-Based Certificate Verification System

1. Introduction

The Blockchain-Based Certificate Verification System is proposed to make academic and professional certificate verification more secure, transparent, and efficient. The system will allow an authorized institution to register certificate details or a cryptographic hash on a blockchain. A verifier can then check whether a presented certificate matches the registered record, helping identify altered or fraudulent documents.

2. Problem Statement

Traditional certificate verification often depends on manual checks, paper records, emails, or direct communication with issuing institutions. These methods can take time and may be vulnerable to forged documents, lost records, and administrative delays. A reliable digital verification process is needed to improve trust and reduce the effort required to validate certificates.

3. Aim and Objectives

The main aim is to design and develop a prototype that uses blockchain technology to support tamper-evident certificate verification. The objectives are to:

- Provide an authorized interface for registering certificate information.
- Generate a unique digital hash for each certificate or its relevant data.
- Record verification references on a blockchain or blockchain test network.
- Allow users or employers to verify a certificate through a unique ID or QR code.
- Display a clear verification result indicating whether the record matches.
- Test the prototype for correctness, usability, and basic security considerations.

4. Proposed Methodology

The project will follow these stages: (1) gather requirements and design the workflow, (2) build a web interface for certificate submission and verification, (3) calculate a cryptographic hash for certificate data or a file, (4) store the hash and a reference identifier on a blockchain test network or suitable local blockchain, (5) generate a QR code or verification link, and (6) test valid, modified, and unregistered certificates. A possible technology stack includes HTML/CSS/JavaScript, a backend such as Python or Node.js, a blockchain development framework, and a database for non-sensitive supporting metadata. The final stack will depend on project resources.

5. Expected Outcomes

The expected outcome is a working academic prototype that can register a certificate reference and verify whether a submitted certificate matches the registered hash. It should reduce reliance on manual checks and demonstrate how blockchain can provide a tamper-evident verification record.

6. Scope and Limitations

The system will focus on a prototype for digitally registering and verifying certificates. Personal information and complete certificate files should not be placed on a public blockchain; only hashes and minimal references should be considered for on-chain storage. The prototype will not independently prove that an issuing institution is legitimate unless issuer authorization and governance are also implemented.

7. Conclusion

This project will demonstrate a practical use of blockchain for improving certificate verification. By combining cryptographic hashes, a verification interface, and a tamper-evident ledger, the proposed system aims to make checking certificate authenticity more convenient and dependable.